

Thermoelastic analysis of functionally graded graphene reinforced rectangular plates based on 3D elasticity

B. Yang · J. Yang · S. Kitipornchai

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Abstract Thermoelastic bending behaviour of novel functionally graded polymer nanocomposite rectangular plate reinforced with graphene nanoplatelets (GPLs) whose weight fraction varies continuously and smoothly along the thickness direction is investigated. The generalized Mian and Spencer method is utilized to obtain the analytical solutions of nanocomposite rectangular plate with two opposite edges simply supported and under a uniformly distributed transverse load and a temperature change. Three GPL distribution patterns are considered. Comparison between the present analytical solutions and those available in literature is carried out to verify the accuracy of our analytical solutions. A parametric study is conducted to examine the effects of GPL's weight fraction, distribution pattern, geometry and size as well as the temperature change and plate boundary conditions on the stress and deformation fields of the nanocomposite plates. Numerical results show that the addition of

GPLs at a very low content can have a significant reinforcing effect on the thermo-mechanical response of the plate.

Keywords Rectangular plate · Functionally graded material · Graphene reinforced composite · Thermo-elasticity · Analytical solutions

1 Introduction

Since the discovery of graphene, huge attentions from both research and engineering communities have been focused on its extraordinary properties and great potential in various applications in science and engineering. With the outstanding mechanical properties of the sp^2 carbon bonding network, graphene is regarded as the strongest and stiffest material ever known. Graphene can be used to construct a single-walled carbon nanotube (SWNT), multi-walled carbon nanotube (MWNT) and graphene platelets (GPLs) which comprise multiple graphene sheets that are stacked together. Due to its high specific surface area and strong nanofiller–matrix adhesion, graphene can serve as excellent thermally conductive nanofillers in composites [1–4]. A large number of experimental and theoretical researches have been conducted in the fabrication of different graphene–polymer nanocomposites and the characterization of their thermo-mechanical properties. Attributed to its two-dimensional geometry, GPLs

B. Yang · J. Yang (✉)
School of Engineering, RMIT University, Bundoora,
VIC 3083, Australia
e-mail: j.yang@rmit.edu.au

B. Yang
Department of Civil Engineering, Zhejiang Sci-Tech
University, Hangzhou 310018, People's Republic of
China

S. Kitipornchai
School of Civil Engineering, University of Queensland,
Brisbane, QLD 4072, Australia

remarkably outperform carbon nanotubes in flexibility, thermal expansion, and electrical and thermal conductivities [5, 6] hence are considered to be better reinforcement than carbon nanotubes for polymer nanocomposites. Many previous studies have reported that graphene–polymer composites at a low GPL content show significantly improved mechanical and thermal properties as compared to their pristine polymer counterparts and carbon nanotube based composites [5, 7, 8].

Quite a few studies have been done to analyze the mechanical behavior of graphene sheets and their composites. For instance, Kitipornchai et al. [9] studied the vibration analysis of multilayered graphene sheets by employing a continuum mechanics model. Wang et al. [7] investigated the thermal expansion of graphene sheets composites by a thermo-mechanical analyzer. Three dimensional vibration analysis of multi-layered graphene sheets embedded in polymer matrix was carried out by Alibeigloo [10] by using nonlocal continuum mechanics. Shahil and Balandin [11] reviewed the thermal properties of graphene and multilayer graphene, and discussed graphene's applications in thermal management in advanced electronics and optoelectronics. Wu and Drzal [12] added GPLs to polyetherimide thermoplastic polymer and measured and modelled the coefficient of thermal expansion and linear thermal expansion coefficient of the polymer composite. On the basis of Eringen's nonlocal elasticity equations and a new two-variable plate theory, Sobhy [13] investigated the free vibration, mechanical buckling and thermal buckling behaviour of multilayered graphene sheets; Sobhy [14] employed Levy-type solution to obtain the bending response of single-layered graphene sheets subjected to a temperature field as well as an external mechanical load.

Functionally graded materials (FGMs) are a novel type of composites of microscopical inhomogeneity which are made of two or more constituent phases whose properties vary spatially and continuously according to a certain distribution of the reinforcement phase. Inspired by the concept of FGMs, functionally graded carbon nanotube reinforced composites (FG-CNTRC) in which CNT distributions within an isotropic matrix are designed specifically to be graded along the desired directions so as to improve mechanical properties of the structures was discussed by Shen [15]. Shen and his co-workers [15–18] further

investigated the nonlinear bending, nonlinear vibration, thermal buckling and postbuckling behaviors of functionally graded nanocomposite plates or shells reinforced by carbon nanotubes. Rafiee et al. [19] studied the nonlinear thermal bifurcation buckling behavior of CNTRC beams with surface-bonded piezoelectric layers. Ke et al. [20] reported a dynamic stability analysis of functionally graded nanocomposite beams reinforced by single-walled carbon nanotubes (SWCNTs) based on Timoshenko beam theory. The dispersion behavior of guide waves in a functionally graded nanocomposite plate reinforced with SWCNTs was investigated by Zhu et al. [21] using the first-order shear deformation plate theory. Based on three-dimensional theory of elasticity, Alibeigloo and Liew [22] presented bending analysis of FG-CNTRC rectangular plate with simply supported edges subjected to thermo-mechanical loadings. Based on the first order shear deformation theory, Zhang et al. [23] studied the vibration of moderately thick FG-CNTRC skew plates by using element-free IMLS-Ritz method. Analytic and finite element models were obtained by Bafekrpour et al. [24] to examine the effects of gradient patterns on the mechanical behaviour of FG-CNTRC beams. More works on the analysis of FG-CNTRC structures can be found in a review paper by Liew et al. [25].

It should be noted that the above-mentioned studies dealing with graphene nanocomposites considered graphene sheets uniformly distributed in the matrix only. The primary objective of this paper is to investigate the thermoelastic bending behaviour of functionally graded graphene reinforced composite rectangular plates subjected to thermal and mechanical loads. The GPL nanofillers are assumed to be either non-uniformly or uniformly dispersed in the matrix to explore the most effective way to make the best use of the potentials of GPL reinforcements. The analytical solution procedure developed by Mian and Spencer [26] for isotropic FGM plates that are tractions free on both upper and bottom surfaces which was later successfully extended to uniformly loaded transversely isotropic FGM plates in our previous work [27] will be further modified to derive the three-dimensional elasticity theory based thermo-mechanical bending solution of functionally graded graphene reinforced composite rectangular plates with two opposite edges simply supported. The plate is subjected to a uniform transverse load in combination

with a temperature field along the thickness direction under one of the three typical thermal boundary conditions. A parametric study is conducted, with a particular focus on the effects of weight fraction, distribution pattern, geometry and size of GPL nanofillers on the deflection and bending stress distributions of the plate under various boundary conditions.

2 FG Polymer nanocomposite reinforced with GPLs

According to the modified Halpin–Tsai micromechanics model, the effective Young’s modulus E_c of the isotropic homogeneous nanocomposite reinforced with uniformly distributed GPLs can be calculated by [5]

$$E_c = \frac{3}{8} \frac{1 + \xi_L \left(\frac{E_{GPL}/E_M - 1}{E_{GPL}/E_M + \xi_L} \right) V_{GPL}}{1 - \left(\frac{E_{GPL}/E_M - 1}{E_{GPL}/E_M + \xi_L} \right) V_{GPL}} E_M + \frac{5}{8} \frac{1 + \xi_W \left(\frac{E_{GPL}/E_M - 1}{E_{GPL}/E_M + \xi_W} \right) V_{GPL}}{1 - \left(\frac{E_{GPL}/E_M - 1}{E_{GPL}/E_M + \xi_W} \right) V_{GPL}} E_M. \quad (1)$$

where L , W and t represent the average length, width and thickness of the GPLs; E_{GPL} and E_M are the moduli of the GPLs and the polymer matrix, respectively; ξ_L and ξ_W are the two geometry parameters describing the effects of GPLs’ shape and size in the longitudinal and transverse directions and can be expressed as

$$\xi_L = 2(L/t), \quad \xi_W = 2(W/t). \quad (2)$$

The volume fraction of the GPL nanofillers V_{GPL} can be calculated from

$$V_{GPL} = \frac{W_{GPL}}{W_{GPL} + \frac{\rho_{GPL}}{\rho_M}(1 - W_{GPL})}. \quad (3)$$

in which W_{GPL} and ρ_{GPL} are GPLs’ weight fraction and mass density; ρ_M is the mass density of the matrix.

According to the rule of mixture [12, 22], the effective Poisson’s ratio ν_c and thermal expansion coefficient α_c of the GPL reinforced nanocomposite can be calculated by

$$\nu_c = V_{GPL}\nu_{GPL} + V_m\nu_m. \quad (4)$$

$$\alpha_c = V_{GPL}\alpha_{GPL} + V_m\alpha_m. \quad (5)$$

where ν_{GPL} , ν_m , α_{GPL} and α_m are Poisson’s ratio and thermal expansion coefficient of GPLs and the matrix, respectively. The volume fraction of the matrix V_m and that of GPLs V_{GPL} are related by

$$V_{GPL} + V_m = 1. \quad (6)$$

The effective thermal conductivity of the GPL reinforced nanocomposite k_c can be derived from the micromechanics theory [3]

$$\frac{k_c}{k_m} = 1 + \frac{V_{GPL}}{3} \times \left[\frac{2}{H + 1/(k_x/k_m - 1)} + \frac{1}{(1 - H)/2 + 1/(k_z/k_m - 1)} \right]. \quad (7)$$

where k_m is the thermal conductivity of the matrix and

$$H = \frac{\ln \left[\left(p + \sqrt{p^2 - 1} \right) p \right]}{\sqrt{(p^2 - 1)^3}} - \frac{1}{p^2 - 1}. \quad (8)$$

$$k_x = \frac{k_g}{2R_k k_g/L + 1}, \quad k_z = \frac{k_g}{2R_k k_g/t + 1}. \quad (9)$$

in which R_k is an average interfacial thermal resistance between the GPLs and matrix, k_g is GPL’s intrinsic thermal conductivity along the in-plane direction, and $p = L/t$.

In functionally graded polymer nanocomposites where the GPLs are non-uniformly dispersed within the polymer matrix, the elastic and thermal constants in Eqs. (1), (4), (5) and (7) are position-dependent as the functions of thickness coordinate of the plate. In the present work, three different GPL distribution models as shown in Fig. 1, including two non-uniform models and one uniform model as a special case, are considered and can be described as

1. Linear GPL distribution with its weight fraction changing from the highest on the top surface to zero at the bottom surface, i.e.,

$$W_{GPL}(z) = \lambda_1 W_{GPL}^0 (1/2 + z/h). \quad (10)$$

2. Nonlinear GPL distribution with its highest weight fraction located at a distance s from the mid-plane, i.e.,

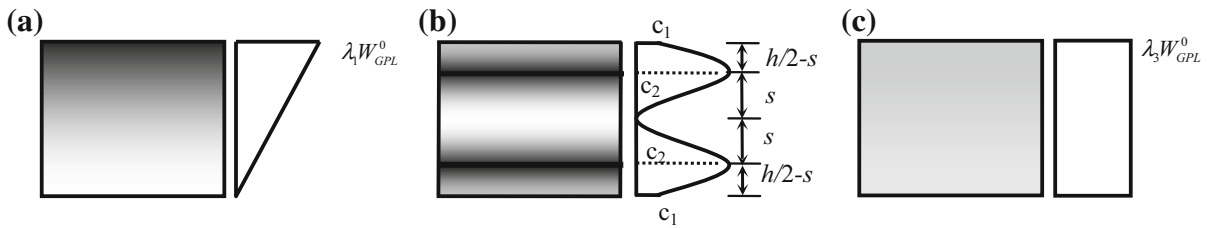


Fig. 1 Schematic representations of compositional gradient configurations of graphene across the thickness of plate: **a** linear, **b** nonlinear, and **c** uniform

$$W_{GPL}(z) = \frac{\lambda_2 W_{GPL}^0 z^2}{h^2 s^2 (4s^2 - h^2)} \times \left[4h^2 z^2 - h^4 + \frac{16}{n} s^2 (s^2 - z^2) \right]. \quad (11)$$

As can be seen in Fig. 1b, $c_1 = c_2/n$ (n is a positive integer) and $c_2 = \lambda_2 W_{GPL}^0$ denote GPL weight fractions at a distance $h/2$ and s from the mid-plane, respectively.

- Uniform GPL distribution with the same weight fraction over the plate thickness, i.e.,

$$W_{GPL}(z) = \lambda_3 W_{GPL}^0. \quad (12)$$

where W_{GPL}^0 is a certain GPL weight fraction, λ_1 , λ_2 and λ_3 are weight fraction indices.

3 Theoretical formulations

3.1 Basic equations

Consider a rectangular plate with length a , width b and height h , defined in the rectangular Cartesian coordinates system $Oxyz$ and subjected to combined mechanical and temperature loading (Fig. 2). Let u , v and w be the displacement components, and σ_{ij} ($i, j = x, y, z$) be the stress components, respectively. Suppose the xy plane coincides with the mid-plane of the plate and the uniformly distributed pressures q_1 and q_2 are applied on the bottom and top surfaces of the plate.

In the absence of body forces, the equations of equilibrium are

$$\sigma_{ij,j} = 0, \quad (13)$$

where a comma denotes partial differentiation with respect to the indicated variable.

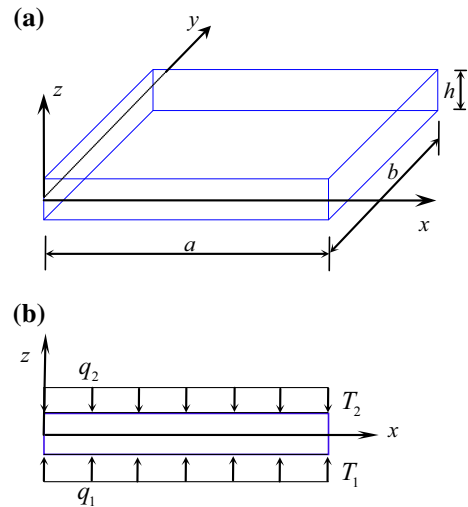


Fig. 2 A thermo-mechanically loaded rectangular plate: **a** 3D view and **b** front view

The linear thermo-elastic stress-strain relationship for transversely isotropic material can be expressed as

$$\begin{aligned} \sigma_{xx} &= c_{11}(u_{,x} - \alpha_1 T) + c_{12}(v_{,y} - \alpha_1 T) + c_{13}(w_{,z} - \alpha_3 T), \\ \sigma_{yy} &= c_{12}(u_{,x} - \alpha_1 T) + c_{11}(v_{,y} - \alpha_1 T) + c_{13}(w_{,z} - \alpha_3 T), \\ \sigma_{zz} &= c_{13}(u_{,x} - \alpha_1 T) + c_{13}(v_{,y} - \alpha_1 T) + c_{33}(w_{,z} - \alpha_3 T), \\ \sigma_{yz} &= c_{44}(w_{,y} + v_{,z}), \quad \sigma_{zx} = c_{44}(u_{,z} + w_{,x}), \\ \sigma_{xy} &= c_{66}(u_{,y} + v_{,x}), \end{aligned} \quad (14)$$

where c_{ij} with $2c_{66} = c_{11} - c_{12}$ are the elastic constants, α_i is the linear thermal expansion coefficients, and T is the temperature rise from the initial stress-free state. Here the z -axis coincides with the material symmetry axis. Note that for the functionally graded GPL/polymer nanocomposites, the above material parameters are the functions of z , i.e., $c_{ij} = c_{ij}(z)$, $\alpha_i = \alpha_i(z)$ and $T = T(z)$. In this paper, the nanocomposites under consideration are isotropic materials hence $c_{11} = c_{33}$, $c_{12} = c_{13}$, $c_{44} = c_{66}$ and $\alpha_1 = \alpha_3$, in which

the elastic constants are related to the effective Young's modulus E_c and Poisson's ratio ν_c by

$$\begin{aligned} c_{12} = c_{13} &= \frac{E_c \nu_c}{(1 + \nu_c)(1 - 2\nu_c)}, \\ c_{11} = c_{33} &= \frac{E_c(1 - \nu_c)}{(1 + \nu_c)(1 - 2\nu_c)}, \\ c_{44} = c_{66} &= \frac{E_c}{2(1 + \nu_c)}. \end{aligned}$$

where E_c and ν_c are defined in Eqs. (1) and (4), respectively.

In what follows we continue to concern the transversely isotropic material. According to Mian and Spencer [26], it is assumed that the solutions of Eqs. (13) and (14) take the form of

$$\begin{aligned} u(x, y, z) &= \bar{u}(x, y) + F\Delta_{,x} + A\bar{w}_{,x} + B\nabla^2\bar{w}_{,x}, \\ v(x, y, z) &= \bar{v}(x, y) + F\Delta_{,y} + A\bar{w}_{,y} + B\nabla^2\bar{w}_{,y}, \\ w(x, y, z) &= \bar{w}(x, y) + G\Delta + C\nabla^2\bar{w} + D, \end{aligned} \quad (15)$$

where A, B, C, D, F and G are the functions of z as well, and

$$\Delta = \Delta(x, y) = \bar{u}_{,x} + \bar{v}_{,y}, \quad \nabla^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2. \quad (16)$$

The functions A, B, C, D, F and G in Eq. (15) can be determined from the traction conditions on the top and bottom surfaces of the plate. The displacement field in Eq. (15) can be obtained once the in-plane displacements $\bar{u}(x, y)$, $\bar{v}(x, y)$ and $\bar{w}(x, y)$ are determined for a given boundary value problem. In what follows, the equations governing these three in-plane displacement functions will be derived.

Substituting Eq. (15) into (14) gives

$$\begin{aligned} \sigma_{xx} &= (c_{12} + c_{13}G')\Delta + (c_{12}A + c_{13}C')\nabla^2\bar{w} \\ &\quad + c_{12}(F\nabla^2\Delta + B\nabla^4\bar{w}) + c_{13}D' \\ &\quad + 2c_{66}(\bar{u}_{,x} + F\Delta_{,xx} + A\bar{w}_{,xx} + B\nabla^2\bar{w}_{,xx}) \\ &\quad - [(c_{11} + c_{12})\alpha_1 + c_{13}\alpha_3]T, \\ \sigma_{yy} &= (c_{12} + c_{13}G')\Delta + (c_{12}A + c_{13}C')\nabla^2\bar{w} \\ &\quad + c_{12}(F\nabla^2\Delta + B\nabla^4\bar{w}) + c_{13}D' \\ &\quad + 2c_{66}(\bar{v}_{,y} + F\Delta_{,yy} + A\bar{w}_{,yy} + B\nabla^2\bar{w}_{,yy}) \\ &\quad - [(c_{11} + c_{12})\alpha_1 + c_{13}\alpha_3]T, \\ \sigma_{zz} &= (c_{13} + c_{33}G')\Delta + (c_{13}A + c_{33}C')\nabla^2\bar{w} \\ &\quad + c_{13}(F\nabla^2\Delta + B\nabla^4\bar{w}) + c_{33}D' - (2c_{13}\alpha_1 + c_{33}\alpha_3)T, \\ \sigma_{yz} &= c_{44}[(A' + 1)\bar{w}_{,y} + (F' + G)\Delta_{,y} + (B' + C)\nabla^2\bar{w}_{,y}], \\ \sigma_{zx} &= c_{44}[(A' + 1)\bar{w}_{,x} + (F' + G)\Delta_{,x} + (B' + C)\nabla^2\bar{w}_{,x}], \\ \sigma_{xy} &= c_{66}(\bar{u}_{,y} + \bar{v}_{,x} + 2F\Delta_{,xy} + 2A\bar{w}_{,xy} + 2B\nabla^2\bar{w}_{,xy}), \end{aligned} \quad (17)$$

where the prime denotes derivative with respect to z .

Substituting Eq. (17) into Eq. (13), one has

$$\begin{aligned} \{c_{11} + c_{13}G' + [c_{44}(F' + G)]'\}\Delta_{,x} - c_{66}\Omega_{,y} \\ + c_{11}F\nabla^2\Delta_{,x} + [c_{44}(A' + 1)]'\bar{w}_{,x} \\ + \{c_{11}A + c_{13}C' + [c_{44}(B' + C)]'\}\nabla^2\bar{w}_{,x} \\ + c_{11}B\nabla^4\bar{w}_{,x} = 0, \end{aligned} \quad (18)$$

$$\begin{aligned} \{c_{11} + c_{13}G' + [c_{44}(F' + G)]'\}\Delta_{,y} + c_{66}\Omega_{,x} \\ + c_{11}F\nabla^2\Delta_{,y} + [c_{44}(A' + 1)]'\bar{w}_{,y} \\ + \{c_{11}A + c_{13}C' + [c_{44}(B' + C)]'\}\nabla^2\bar{w}_{,y} \\ + c_{11}B\nabla^4\bar{w}_{,y} = 0, \end{aligned} \quad (19)$$

$$\begin{aligned} (c_{13} + c_{33}G')'\Delta + [c_{44}(F' + G) + (c_{13}F)']\nabla^2\Delta \\ + [c_{44}(A' + 1) + (c_{13}A + c_{33}C')']\nabla^2\bar{w} \\ + [c_{44}(B' + C) + (c_{13}B)']\nabla^4\bar{w} \\ + [c_{33}D' - (2c_{13}\alpha_1 + c_{33}\alpha_3)T]' = 0. \end{aligned} \quad (20)$$

By setting

$$[c_{44}(A' + 1)]' = 0, \quad (21)$$

$$(c_{13} + c_{33}G')' = 0, \quad (22)$$

$$c_{11} + c_{13}G' + [c_{44}(F' + G)]' = c_{66}\kappa_1, \quad (23)$$

$$c_{11}A + c_{13}C' + [c_{44}(B' + C)]' = c_{66}\kappa_2. \quad (24)$$

$$c_{44}(A' + 1) + (c_{13}A + c_{33}C')' = 0, \quad (25)$$

where κ_1 and κ_2 are arbitrary constants, Eqs. (18)–(20) become

$$\begin{aligned} c_{66}(\kappa_1\Delta_{,x} - \Omega_{,y} + \kappa_2\nabla^2\bar{w}_{,x}) \\ + c_{11}(F\nabla^2\Delta + B\nabla^4\bar{w})_{,x} = 0, \end{aligned} \quad (26)$$

$$\begin{aligned} c_{66}(\kappa_1\Delta_{,y} + \Omega_{,x} + \kappa_2\nabla^2\bar{w}_{,y}) \\ + c_{11}(F\nabla^2\Delta + B\nabla^4\bar{w})_{,y} = 0, \end{aligned} \quad (27)$$

$$\begin{aligned} H\nabla^2\Delta + I\nabla^4\bar{w} \\ + [c_{33}D' - (2c_{13}\alpha_1 + c_{33}\alpha_3)T]' = 0, \end{aligned} \quad (28)$$

in which

$$\begin{aligned}\Omega &= \bar{v}_{,x} - \bar{u}_{,y}, \quad H = c_{44}(F' + G) + (c_{13}F)', \\ I &= c_{44}(B' + C) + (c_{13}B)'. \end{aligned} \quad (29)$$

If H and I are not zero simultaneously, it follows from Eq. (28) that

$$\nabla^2 \Delta = \kappa_3, \quad \nabla^4 \bar{w} = \kappa_4. \quad (30)$$

where κ_3 and κ_4 are arbitrary constants. Equations (26)–(28) then become

$$(\kappa_1 \Delta + \kappa_2 \nabla^2 \bar{w})_{,x} - \Omega_{,y} = 0, \quad (31)$$

$$(\kappa_1 \Delta + \kappa_2 \nabla^2 \bar{w})_{,y} + \Omega_{,x} = 0, \quad (32)$$

$$H\kappa_3 + I\kappa_4 + [c_{33}D' - (2c_{13}\alpha_1 + c_{33}\alpha_3)T]' = 0. \quad (33)$$

By virtue of Eq. (30), it can be obtained from Eqs. (31) and (32) that

$$\kappa_1 \kappa_3 + \kappa_2 \kappa_4 = 0. \quad (34)$$

Equations (30)₂, (31) and (32) constitute the governing equations for in-plane displacement functions $\bar{u}(x, y)$, $\bar{v}(x, y)$ and $\bar{w}(x, y)$ for the thermo-elasticity problem under consideration.

3.2 Temperature field $T(z)$

This paper considers a one-dimensional steady state temperature field $T(z)$ varying along the thickness direction due to heat conduction induced by different temperatures at the top and bottom surfaces of the plate whose boundaries at $x = 0, a$ and $y = 0, b$ are assumed to be adiabatic. According to the Fourier law of heat conduction [28], the thermal flux q_z in the z -direction is

$$q_z = -k(z) \frac{dT}{dz}. \quad (35)$$

where $k = k(z)$ is the thermal conductivity coefficient. The temperature field can be derived from the following equation

$$-\frac{d}{dz} \left[k(z) \frac{dT}{dz} \right] = 0. \quad (36)$$

with the solution being expressed as

$$T(z) = C_1 \int_{-h/2}^z \frac{d\eta}{k(\eta)} + C_2. \quad (37)$$

where integral constants C_1 and C_2 are to be determined by the thermal conditions on the top and bottom surfaces of the plate.

There are three kinds of linear thermal boundary conditions [28, 29], including the Dirichlet boundary conditions with the prescribed temperatures, Neumann boundary conditions with the prescribed thermal flux and Robin boundary conditions satisfying Newton's law of cooling. Totally, there are nine different combinations of thermal boundary conditions on the top and bottom surfaces of the plate. The following three typical boundary conditions are described, as an example

1. The Dirichlet conditions used at $z = \pm h/2$, namely

$$T|_{z=-h/2} = T_1, \quad T|_{z=h/2} = T_2. \quad (38)$$

where T_1 and T_2 are known constants.

It follows from Eqs. (37) and (38) that

$$C_1 = (T_2 - T_1) \left/ \int_{-h/2}^{h/2} \frac{d\eta}{k(\eta)} \right., \quad C_2 = T_1. \quad (39)$$

2. The Dirichlet condition used at $z = h/2$ and the Neumann condition employed at $z = -h/2$, namely

$$q_z|_{z=-h/2} = \Theta_1, \quad T|_{z=h/2} = T_2. \quad (40)$$

where Θ_1 is a known constant.

It follows from Eqs. (35), (37) and (40) that

$$C_1 = -\Theta_1, \quad C_2 = T_2 + \Theta_1 \int_{-h/2}^{h/2} \frac{d\eta}{k(\eta)}. \quad (41)$$

3. The Dirichlet condition used at $z = h/2$ and the Newton's law of cooling condition employed at $z = -h/2$, namely

$$(T - \beta q_z)|_{z=-h/2} = \Xi_1, \quad T|_{z=h/2} = T_2. \quad (42)$$

where β and Ξ_1 are known constants.

It follows from Eqs. (35), (37) and (42) that

$$\begin{aligned} C_1 &= (\Xi_1 - T_2) \left/ \left[\beta - \int_{-h/2}^{h/2} \frac{d\eta}{k(\eta)} \right] \right., \\ C_2 &= \left(\beta T_2 - \Xi_1 \int_{-h/2}^{h/2} \frac{d\eta}{k(\eta)} \right) \left/ \left[\beta - \int_{-h/2}^{h/2} \frac{d\eta}{k(\eta)} \right] \right. \end{aligned} \quad (43)$$

As a result, the temperature field in Eq. (37) can be completely determined when the integral constants C_1 and C_2 are determined from Eqs. (39), (41) or (43).

3.3 Determination of functions

$A(z)$, $B(z)$, $C(z)$, $D(z)$, $F(z)$ and $G(z)$

In this section, the unknown functions $A(z)$, $B(z)$, $C(z)$, $D(z)$, $F(z)$ and $G(z)$ will be determined by utilizing the traction stress conditions $\sigma_{xz} = \sigma_{yz} = 0$ at $z = \pm h/2$, $\sigma_z = -q_1$ at $z = -h/2$, and $\sigma_z = -q_2$ at $z = h/2$.

By substituting the expressions of σ_{xz} , σ_{yz} and σ_z in Eq. (17) into the above traction stress boundary conditions at $z = \pm h/2$, we obtain

$$A'(\pm h/2) + 1 = 0. \quad (44)$$

$$c_{13}(\pm h/2) + c_{33}(\pm h/2)G'(\pm h/2) = 0. \quad (45)$$

$$F'(\pm h/2) + G(\pm h/2) = 0. \quad (46)$$

$$B'(\pm h/2) + C(\pm h/2) = 0. \quad (47)$$

$$c_{13}(\pm h/2)A(\pm h/2) + c_{33}(\pm h/2)C'(\pm h/2) = 0. \quad (48)$$

$$\begin{aligned} c_{13}(-h/2)[\kappa_3 F(-h/2) + \kappa_4 B(-h/2)] \\ + c_{33}(-h/2)D'(-h/2) \\ - [2c_{13}(-h/2)\alpha_1(-h/2) + c_{33}(-h/2) \\ \alpha_3(-h/2)]T(-h/2) = -q_1. \end{aligned} \quad (49)$$

$$\begin{aligned} c_{13}(h/2)[\kappa_3 F(h/2) + \kappa_4 B(h/2)] + c_{33}(h/2)D'(h/2) \\ - [2c_{13}(h/2)\alpha_1(h/2) + c_{33}(h/2)\alpha_3(h/2)]T(h/2) \\ = -q_2. \end{aligned} \quad (50)$$

Given that $\bar{u}(x, y)$, $\bar{v}(x, y)$, and $\bar{w}(x, y)$ are mid-plane displacements, namely

$$\begin{aligned} \bar{u}(x, y) = u(x, y, 0), \quad \bar{v}(x, y) = v(x, y, 0), \\ \bar{w}(x, y) = w(x, y, 0). \end{aligned} \quad (51)$$

We can find from Eq. (15) that

$$\begin{aligned} A(0) = 0, \quad B(0) = 0, \quad C(0) = 0, \quad D(0) = 0, \\ F(0) = 0, \quad G(0) = 0. \end{aligned} \quad (52)$$

The expressions of $A(z)$, $B(z)$, $C(z)$, $D(z)$, $F(z)$ and $G(z)$ can then be determined by integrating Eqs. (21)–

(25) and (33) and making use of Eqs. (44)–(50), (34) and (52) as follows

$$A(z) = -z. \quad (53)$$

$$B(z) = \kappa_2 f_6(z) + f_4(z) - g_1^1(z) - (z + h/2)C_0 + B_0. \quad (54)$$

$$C(z) = g_1^0(z) + C_0. \quad (55)$$

$$D(z) = \kappa_3 f_7(z) - \kappa_4 f_8(z) - q_1 f_3(z) + f_9(z) + D_0 \quad (56)$$

$$F(z) = \kappa_1 f_6(z) - f_5(z) + g_0^1(z) - (z + h/2)G_0 + F_0. \quad (57)$$

$$G(z) = -g_0^0(z) + G_0. \quad (58)$$

where related functions and constants are listed in “Appendix 1”.

3.4 Resultant forces

This section gives the resultant forces expressed in terms of mid-plane displacements which are needed for the cylindrical boundary conditions of the plate. By virtue of Eq. (17), the resultant forces and bending moments can be obtained, respectively

$$\begin{aligned} N_x = \int_{-h/2}^{h/2} \sigma_x dz = N_1 \bar{u}_{,x} + N_2 \bar{v}_{,y} + N_3 \bar{w}_{,xx} \\ + N_4 \bar{w}_{,yy} + N_5 \Delta_{,xx} + N_6 \Delta_{,yy} \\ + N_7 \nabla^2 \bar{w}_{,xx} + N_8 \nabla^2 \bar{w}_{,yy} + N_0, \end{aligned} \quad (59)$$

$$\begin{aligned} N_y = \int_{-h/2}^{h/2} \sigma_y dz = N_2 \bar{u}_{,x} + N_1 \bar{v}_{,y} + N_4 \bar{w}_{,xx} \\ + N_3 \bar{w}_{,yy} + N_6 \Delta_{,xx} + N_5 \Delta_{,yy} \\ + N_8 \nabla^2 \bar{w}_{,xx} + N_7 \nabla^2 \bar{w}_{,yy} + N_0, \end{aligned} \quad (60)$$

$$\begin{aligned} N_{xy} = \int_{-h/2}^{h/2} \tau_{xy} dz \\ = T_1 (\bar{u}_{,y} + \bar{v}_{,x}) + 2T_2 \bar{w}_{,xy} + 2T_3 \Delta_{,xy} + 2T_4 \nabla^2 \bar{w}_{,xy} \end{aligned} \quad (61)$$

$$\begin{aligned}
 M_x = \int_{-h/2}^{h/2} \sigma_x z dz &= M_1 \bar{u}_{,x} + M_2 \bar{v}_{,y} + M_3 \bar{w}_{,xx} \\
 &+ M_4 \bar{w}_{,yy} + M_5 \Delta_{,xx} + M_6 \Delta_{,yy} \\
 &+ M_7 \nabla^2 \bar{w}_{,xx} + M_8 \nabla^2 \bar{w}_{,yy} + M_0,
 \end{aligned} \quad (62)$$

$$\begin{aligned}
 M_y = \int_{-h/2}^{h/2} \sigma_y z dz &= M_2 \bar{u}_{,x} + M_1 \bar{v}_{,y} + M_4 \bar{w}_{,xx} \\
 &+ M_3 \bar{w}_{,yy} + M_6 \Delta_{,xx} + M_5 \Delta_{,yy} \\
 &+ M_8 \nabla^2 \bar{w}_{,xx} + M_7 \nabla^2 \bar{w}_{,yy} + M_0,
 \end{aligned} \quad (63)$$

$$\begin{aligned}
 M_{xy} &= \int_{-h/2}^{h/2} \tau_{xy} z dz \\
 &= S_1 (\bar{u}_{,y} + \bar{v}_{,x}) + 2S_2 \bar{w}_{,xy} + 2S_3 \Delta_{,xy} + 2S_4 \nabla^2 \bar{w}_{,xy}
 \end{aligned} \quad (64)$$

$$Q_x = \int_{-h/2}^{h/2} \tau_{xz} dz = Q_1 \Delta_{,x} + Q_2 \nabla^2 \bar{w}_{,x}, \quad (65)$$

$$Q_y = \int_{-h/2}^{h/2} \tau_{yz} dz = Q_1 \Delta_{,y} + Q_2 \nabla^2 \bar{w}_{,y} \quad (66)$$

where $N_0, \dots, N_8, M_0, \dots, M_8, T_1, \dots, T_4, S_1, \dots, S_4, Q_1, Q_2$ are integral constants given in “Appendix 2”.

3.5 Rectangular plate with two opposite edges simply supported

In this paper, a rectangular plate with a pair of opposite edges simply supported at $x = 0, a$ is considered. Its mid-plane displacements are assumed to be in the form of

$$\bar{u} = \sum_{m=1}^{\infty} u_m(y) \cos \alpha_m x + A_1 x, \quad (67)$$

$$\bar{v} = \sum_{m=1}^{\infty} v_m(y) \sin \alpha_m x, \quad (68)$$

$$\bar{w} = \sum_{m=1}^{\infty} w_m(y) \sin \alpha_m x + A_2 x(x-a) + \frac{1}{24} \kappa_4 x^2 (x-a)^2, \quad (69)$$

where $\alpha_m = m\pi/a$, and A_1 and A_2 are undetermined constants.

Substituting Eqs. (67)–(69) into Eqs. (30)₂, (31) and (32) leads to

$$\frac{d^4 w_m}{dy^4} - 2\alpha_m^2 \frac{d^2 w_m}{dy^2} + \alpha_m^4 w_m = 0, \quad (70)$$

$$\begin{aligned}
 &\left(\frac{d^2}{dy^2} - \kappa_1 \alpha_m^2 \right) u_m + (\kappa_1 - 1) \alpha_m \frac{dv_m}{dy} \\
 &+ \kappa_2 \alpha_m \left(\frac{d^2}{dy^2} - \alpha_m^2 \right) w_m - \frac{2\kappa_2 \kappa_4}{a \alpha_m^2} [1 - (-1)^m] = 0,
 \end{aligned} \quad (71)$$

$$\begin{aligned}
 &-(\kappa_1 - 1) \alpha_m \frac{du_m}{dy} + \left(\kappa_1 \frac{d^2}{dy^2} - \alpha_m^2 \right) v_m \\
 &+ \kappa_2 \left(\frac{d^2}{dy^2} - \alpha_m^2 \right) \frac{dw_m}{dy} = 0,
 \end{aligned} \quad (72)$$

The solution of Eq. (70) is

$$w_m = \sum_{i=1}^4 F_{mi} g_{mi}(y), \quad (73)$$

where $F_{mi} (i = 1, 2, 3, 4)$ are integral constants, and

$$\begin{aligned}
 g_{m1}(y) &= e^{-\alpha_m(b-y)}, & g_{m2}(y) &= e^{-\alpha_m y}, \\
 g_{m3}(y) &= ye^{-\alpha_m(b-y)}, & g_{m4}(y) &= ye^{-\alpha_m y}.
 \end{aligned} \quad (74)$$

u_m and v_m can be obtained by substituting Eq. (73) into Eqs. (71) and (72) as follows

$$\begin{aligned}
 u_m &= \alpha_m^2 (\kappa_1 - 1) [G_{m1} g_{m1}(y) + G_{m2} g_{m2}(y)] \\
 &+ \alpha_m G_{m3} [2\kappa_1 g_{m1}(y) + \alpha_m (\kappa_1 - 1) g_{m3}(y)] \\
 &+ \alpha_m G_{m4} [-2\kappa_1 g_{m2}(y) + \alpha_m (\kappa_1 - 1) g_{m4}(y)] \\
 &+ \frac{2\kappa_2}{\kappa_1 - 1} [F_{m3} g_{m1}(y) - F_{m4} g_{m2}(y)] \\
 &+ \frac{2\kappa_3}{a \alpha_m^4} [1 - (-1)^m], \\
 v_m &= \alpha_m^2 (\kappa_1 - 1) [G_{m1} g_{m1}(y) - G_{m2} g_{m2}(y)] \\
 &+ \alpha_m G_{m3} (\kappa_1 - 1) [g_{m1}(y) + \alpha_m g_{m3}(y)] \\
 &+ \alpha_m G_{m4} (\kappa_1 - 1) [g_{m2}(y) - \alpha_m g_{m4}(y)],
 \end{aligned} \quad (75)$$

where $G_{mi} (i = 1, 2, 3, 4)$ are integral constants. The eight unknown constants F_{mi} and $G_{mi} (i = 1, 2, 3, 4)$ can be determined from the boundary conditions at $y = 0$ and $y = b$.

Substituting Eqs. (67)–(69) into Eqs. (59)–(66) yields

$$N_x = \left(N_7 + \frac{a^2}{12} N_3 \right) \kappa_4 + N_0 + N_1 A_1 + 2N_3 A_2 + \sum_{m=1}^{\infty} \tilde{N}_{xm}(y) \sin \alpha_m x + \frac{1}{2} N_3 \kappa_4 x(x-a) \quad (76)$$

$$N_y = \left(N_8 + \frac{a^2}{12} N_4 \right) \kappa_4 + N_0 + N_2 A_1 + 2N_4 A_2 + \sum_{m=1}^{\infty} \tilde{N}_{ym}(y) \sin \alpha_m x + \frac{1}{2} N_4 \kappa_4 x(x-a) \quad (77)$$

$$N_{xy} = \sum_{m=1}^{\infty} \tilde{N}_{xym}(y) \cos \alpha_m x \quad (78)$$

$$M_x = \left(M_7 + \frac{a^2}{12} M_3 \right) \kappa_4 + M_0 + M_1 A_1 + 2M_3 A_2 + \sum_{m=1}^{\infty} \tilde{M}_{xm}(y) \sin \alpha_m x + \frac{\kappa_4}{2} M_3 x(x-a) \quad (79)$$

$$M_y = \left(M_8 + \frac{a^2}{12} M_4 \right) \kappa_4 + M_0 + M_2 A_1 + 2M_4 A_2 + \sum_{m=1}^{\infty} \tilde{M}_{ym}(y) \sin \alpha_m x + \frac{\kappa_4}{2} M_4 x(x-a) \quad (80)$$

$$M_{xy} = \sum_{m=1}^{\infty} \tilde{M}_{xym}(y) \cos \alpha_m x \quad (81)$$

$$Q_x = \sum_{m=1}^{\infty} Q_{xm}(y) \cos \alpha_m x + \frac{1}{2} Q_2 \kappa_4 (2x-a) \quad (82)$$

$$Q_y = \sum_{m=1}^{\infty} Q_{ym}(y) \sin \alpha_m x \quad (83)$$

where the expressions of $\tilde{N}_{xm}(y)$, $\tilde{N}_{ym}(y)$, $\tilde{N}_{xym}(y)$, $\tilde{M}_{xm}(y)$, $\tilde{M}_{ym}(y)$, $\tilde{M}_{xym}(y)$, $Q_{xm}(y)$ and $Q_{ym}(y)$ are given in “Appendix 3”.

Suppose

$$\begin{aligned} \left(N_7 + \frac{a^2}{12} N_3 \right) \kappa_4 + N_0 + N_1 A_1 + 2N_3 A_2 &= 0, \\ \left(M_7 + \frac{a^2}{12} M_3 \right) \kappa_4 + M_0 + M_1 A_1 + 2M_3 A_2 &= 0. \end{aligned} \quad (84)$$

to solve for A_1 and A_2 . It can be found from Eqs. (68), (69), (76), (79) and (84) that the simple support boundary conditions at $x = 0$ and a are all satisfied.

The other two edges at $y = 0$ or $y = b$ can be simply supported (S), clamped (C) or free (F), namely

$$\text{S: } \bar{u} = \bar{w} = 0, \quad N_y = 0, \quad M_y = 0. \quad (85)$$

$$\text{C: } \bar{u} = \bar{v} = \bar{w} = 0, \quad \bar{w}_{,y} = 0. \quad (86)$$

$$\text{F: } N_y = N_{xy} = 0, \quad M_y = 0, \quad Q_y + M_{xy,x} = 0. \quad (87)$$

By virtue of Eqs. (85)–(87) and the orthogonal properties of trigonometric functions, one can obtain eight equations from which the eight integral constants F_{mi} and G_{mi} ($i = 1, 2, 3, 4$) for each m can be determined. Finally, the unknown functions u_m , v_m , w_m , the three dimensional displacement components, and the associated stress fields can be determined successively.

4 Numerical results and discussion

4.1 Validation example

In order to validate the present analytical solutions, a numerical example is given for SSSS and SCSC isotropic and homogeneous rectangular plates subjected to a linearly varying temperature field $T(z) = z\Delta T_0/h$. Dimensionless central deflections $W = 10hw/(\alpha\Delta T_0 a^2)$ at $z = h/2$ for plates with different aspect ratio $a/b = 1.0, 1.5$, and 2.0 are listed in Table 1. The present solution converges rapidly and approaches to the final results by taking $m = 25$ in the series solutions (67)–(69). Our results are in excellent agreement with those by Das and Rath [30] based on Reissner’s variational principle.

4.2 Parametric study

A parametric study is carried out to investigate the influences of the weight fraction, distribution pattern, geometry and size of GPL nanofillers and boundary conditions on the stress and displacement fields of the functionally graded GPL/polymer plates. Epoxy resin is selected as the polymer matrix. Unless stated otherwise, the plate is simply supported with the

Table 1 Dimensionless central displacement $W = 10hw/(\alpha\Delta T_0a^2)$ at $z = h/2$

m	SSSS			CSCS		
	$a/b = 1$	$a/b = 1.5$	$a/b = 2$	$a/b = 1$	$a/b = 1.5$	$a/b = 2$
10	0.9817	0.5934	0.3767	0.2354	0.0498	0.0133
25	0.9810	0.5926	0.3759	0.2346	0.0490	0.0125
50	0.9810	0.5926	0.3759	0.2346	0.0490	0.0125
Ref. [30]	0.967	0.589	0.375	0.235	0.050	0.0114

following parameters: $L = 2.5 \mu\text{m}$, $W = 1.5 \mu\text{m}$, $t = 1.5 \text{ nm}$, $E_{GPL} = 1.01 \text{ TPa}$, $E_M = 2.85 \text{ GPa}$, $\rho_M = 1.2 \times 10^3 \text{ kg/m}^3$, $\rho_{GPL} = 1.06 \times 10^3 \text{ kg/m}^3$ [5]; $k_M = 0.2 \text{ W/mK}$, $k_g = 2000 \text{ W/mK}$, $R_k = 10^{-8} \text{ m}^2\text{K/W}$ [3]; $\alpha_M = 8.2 \times 10^{-5}/\text{K}$ [7]; $\nu_{GPL} = 0.006$, $\alpha_{GPL} = 2.35 \times 10^{-5}/\text{K}$ [12]; $\nu_M = 0.34$, $W_{GPL}^0 = 1\%$, $a = 1 \text{ m}$, $b = 1 \text{ m}$, $h = 0.1 \text{ m}$, $q_1 = 10^3 \text{ N/m}^2$. The temperature is held at $T_1 = 280 \text{ K}$ at the bottom surface of the plate and $T_2 = 370 \text{ K}$ at the top surface, respectively. For GPL nonlinear distribution model, $n = 4$ and $s = 0.45h$ are used.

To facilitate the parametric study, the total weight fraction of GPL nanofillers is assumed to be the same in all of the following examples. Table 2 list the relationship between the gradient index λ_i and GPL total weight fraction W_{GPL}^* for three distribution patterns.

Figure 3 depicts the dimensionless through-thickness temperature distribution T/T_1 which is found to be different corresponding to three different GPL distribution patterns. The result for the pristine epoxy plate is also displayed for comparison. Due to their isotropic homogenous nature, the temperature field for both the pristine epoxy plate and the polymer nanocomposite plate with uniformly distributed GPLs are identical and linearly vary across the thickness. This is because for homogeneous plates, the heat conduction induced temperature field is independent on the thermal conductivity coefficient, as indicated by Eq. (36).

Table 2 Relationship between gradient index and total content of GPLs

λ_1 (linear)	λ_2 (nonlinear)	λ_3 (uniform)	W_{GPL}^* (%)
0	0	0	0
0.67	0.43	0.33	0.33
2	1.29	1	1

Figure 4 shows the through-thickness distribution of the dimensionless elastic constant C_{11}/E_m which, as can be seen, are completely consistent with GPL distribution patterns. At a given total wt% of GPLs, the elastic constant remains constant for uniform GPL distribution, increases linearly from the bottom to top surfaces for linear GPL distribution and change nonlinearly when GPLs are nonlinearly dispersed according to Eq. (11). As expected, an increase in GPL wt% leads to an remarkably increased elastic constant which reaches its maximum value at the top surface for linear distribution pattern while for nonlinear distribution pattern, its maximum values occur at a distance of $0.5h-s$ from both top and bottom surfaces. Among the three GPL distribution patterns considered, the linear one has the highest maximum elastic constant which is much bigger than that in uniform distribution. The significant reinforcing effect is clearly demonstrated in this figure as well. For example, the elastic constant C_{11}/E_m of the nanocomposite with uniform distribution is about 2.1 times and 4.4 times that of the pristine epoxy resin plate when only 0.33% and 1.0% of GPLs are added into the epoxy matrix, respectively.

Table 3 lists dimensionless deflections $w(a/2, b/2, h/2)/h \times 10^4$ and normal stress $\sigma_x(a/2, b/2, h/2)/E_m \times 10^5$ of a simply supported plate subjected to a uniform transverse load. The plate becomes stronger and stiffer as the total amount of GPLs increases from 0.0% which corresponds to a pure epoxy plate to 1.0%, thus significantly reducing the bending deflection by 72% for linear GPL distribution, by 81% for nonlinear GPL distribution, and by 78% for uniform GPL distribution. Compared with other GPL distribution patterns, the nonlinear distribution yields both the smallest deflection and the lowest normal bending stress at the top surface of the plate. It should also be noted that as GPL loading increases, the normal stress increases when GPLs are linearly distributed, remains

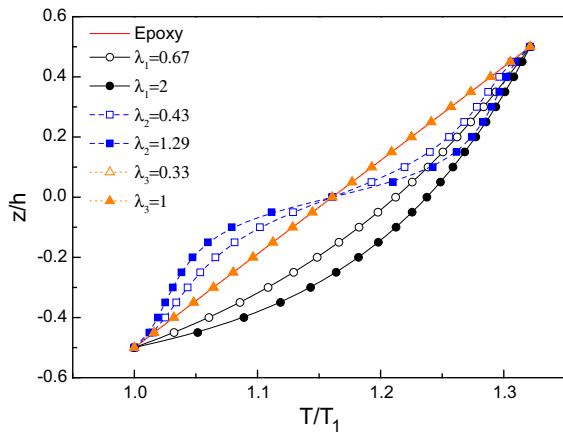


Fig. 3 Dimensionless temperature T/T_1 distribution along the thickness direction

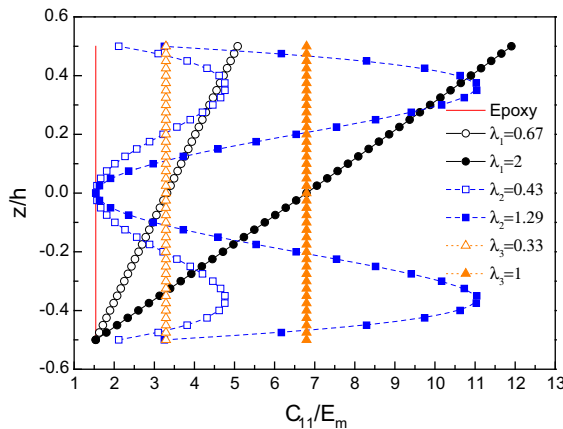


Fig. 4 Distribution of dimensionless elastic constant C_{11}/E_m along the thickness direction

almost unchanged when GPLs are uniformly dispersed, but is decreased when GPLs are nonlinearly embedded in the matrix.

Table 4 gives the dimensionless deflection $w(a/2, b/2, h/2)/h \times 10$ and normal stress $\sigma_x(a/2, b/2, h/2)/E_m \times 10$ of the plate with various boundary conditions under heat conduction with $T_1 = 280$ K at the bottom surface and $T_2 = 370$ K at the top surface. Unlike the mechanical bending discussed in Table 3, the nonlinear GPL distribution leads to the maximum thermally induced deflection and the lowest normal stress while linear GPL distribution does the opposite.

In what follows, numerical results are presented in graphical form for the dimensionless deflection and normal stress at the centre ($x = a/2$, $y = b/2$) of the plate in which GPLs with a total weight fraction of 0.33% and 1.0% are distributed, unless stated otherwise.

Figure 5 plots the through-thickness distribution of dimensionless deflection and normal stress of the nanocomposite plate under a uniform transverse load. The dimensionless deflection $w/h \times 10^4$ is almost constant along the plate thickness except for pure epoxy plate whose deflection is slightly nonlinear, a phenomenon that can only be captured by three-dimensional elasticity solution. Significant reinforcing effect on the plate stiffness can be observed in Fig. 5a where the deflection of a GPL reinforced plate is seen to be much smaller than that of the pristine epoxy plate. The deflection decreases considerably as GPL weight fraction increases. At the same GPL weight fraction, the symmetric and nonlinear GPL distribution patterns provides the best reinforcement among the three distribution patterns considered as it yields the lowest deflection. This is because the bending rigidity of the plate is most improved by the nonlinear distribution pattern which makes the best use of the high modulus of GPLs. The dimensionless normal stress distribution $\sigma_x/E_m \times 10^5$ is more complicated than the deflection. The normal stress changes linearly and symmetrically along the thickness of the pure epoxy plate, exhibits a parabolic and un-symmetric

Table 3 Dimensionless deflection and stress of plate subjected to uniform load

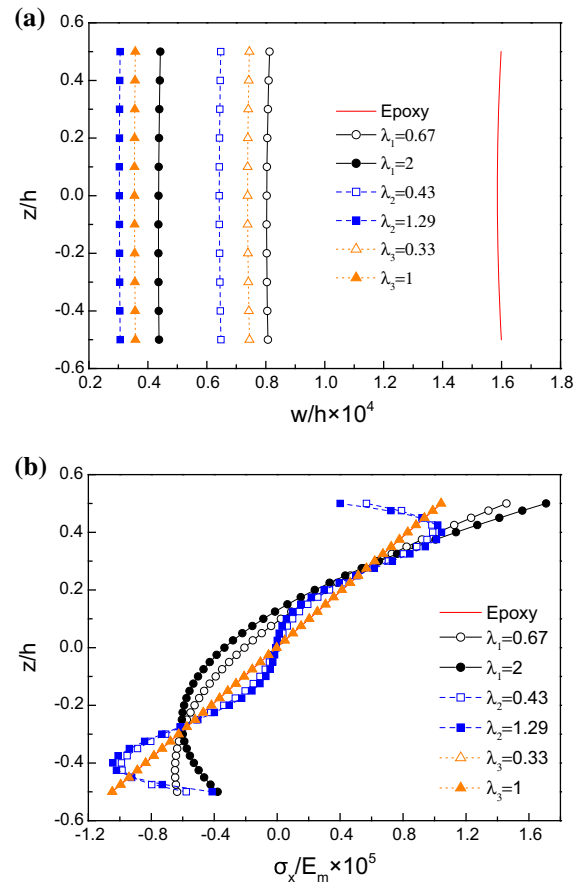
W_{GPL}^* (%)	Distribution case	$w(a/2, b/2, h/2)/h \times 10^4$	$\sigma_x(a/2, b/2, h/2)/E_m \times 10^5$
0	Pure epoxy resin	1.5981	1.0438
0.33	Linear	0.8128	1.4569
	Nonlinear	0.6474	0.5675
	Uniform	0.7443	1.0429
1	Linear	0.4423	1.7090
	Nonlinear	0.3057	0.4004
	Uniform	0.3575	1.0409

Table 4 Dimensionless deflection and stress of plate subjected to thermal load

B.C.	$w(a/2, b/2, h/2)/h \times 10$			$\sigma_x(a/2, b/2, h/2)/E_m \times 10$		
	$\lambda_1 = 2$	$\lambda_2 = 1.29$	$\lambda_3 = 1$	$\lambda_1 = 2$	$\lambda_2 = 1.29$	$\lambda_3 = 1$
SSSS	0.7661	1.1549	1.0051	5.9435	1.6800	3.4436
SSCC	0.4229	0.5086	0.4702	6.4137	1.7412	3.6039
SSFF	1.0687	1.7017	1.4454	6.5754	1.8702	3.8232
SSSC	0.5559	0.7646	0.6811	6.0957	1.6854	3.4726
SSSF	0.8928	1.3841	1.1886	6.3014	1.7832	3.6517
SSCF	0.6009	0.8520	0.7462	6.3742	1.7614	3.6270

variation for the plate with linear GPL distribution, and vary nonlinearly but symmetrically when GPLs are nonlinearly dispersed. It is worth of noting that by moving the maximum GPL concentration away from the top and bottom surfaces by a short distance in the nonlinear GPL distribution, not only the deflection and the normal stresses at both surfaces will become the lowest, the maximum normal stress obtained in this GPL distribution is also much less than that produced by linear GPL distribution, indicating that distributing GPLs in such a nonlinear pattern is the most effective way to reduce both bending deflection and normal stress while reducing GPL concentration linearly from the top to bottom surfaces does significantly bring down the deflection but also results in the biggest maximum normal stress which is not desirable.

In order to discuss the influence of the size and geometry of GPL nanofillers on the bending behavior of the plate, Fig. 6 shows the dimensionless deflection $w/h \times 10^4$ and normal stress $\sigma_x/E_m \times 10^5$ with varying aspect ratio W/L and thickness ratio t/L of GPLs subjected to a uniform transverse load. The GPL length L is kept constant. $W = L$ and $W = L/2$ correspond to square and rectangular shaped GPLs, respectively, with the surface area of the squared GPLs being twice that of the rectangular ones. Moreover, a bigger t/L ratio virtually presents more graphene monolayers used in GPLs. Results show that the deflection can be most effectively reduced by using squared GPLs with fewer single graphene layers. This is because such GPL nanofillers have higher stiffness and better load transfer capability through bigger surface area which can most significantly strengthen the plate stiffness. Figure 6b shows that the normal stress behaves differently for different GPL distribution patterns. It increases (drops) when GPLs that are nonlinearly (linearly) dispersed in the matrix are


Fig. 5 Effect of GPL weight fraction on through-thickness distribution of **a** dimensionless deflection; and **b** normal stress of the plate subjected to uniform load

composed of more single graphene layers. A reduced W/L ratio leads to a slightly increased (reduced) normal stress for nonlinear (linear) GPL distribution pattern. When GPLs are uniformly distributed, the effects of both GPL size and geometry on the normal bending stress are almost negligible.

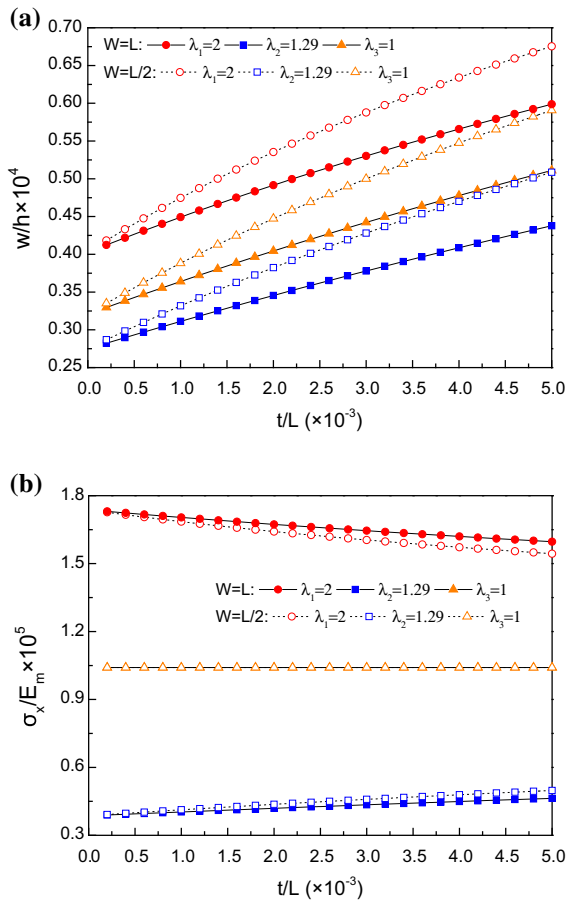


Fig. 6 Effects of GPL geometry and size on the bending behavior of the plate under a uniform load: **a** dimensionless deflection, **b** dimensionless normal stress

Figure 7 presents the dimensionless deflection $w/h \times 10$ and normal stress $\sigma_x/E_m \times 10$ along the thickness direction of the plate subjected to one-dimensional heat conduction induced temperature field where temperature difference $\Delta T = T_2 - T_1$ and $T_1 = 280\text{K}$. In what follows, unless stated otherwise, nonlinear distribution pattern is only considered. It is observed that a higher temperature difference ΔT results in greatly increased bending deflection and larger normal stress at some distance below the top surface where the GPL concentration reaches its peak value. Below the mid-plane, the bending normal stress is almost not affected by the temperature change.

Figure 8 displays the through-thickness distributions of dimensionless deflection $w/h \times 10$ and normal stress $\sigma_x/E_m \times 10$ of the plate with different boundary conditions subjected to heat conduction with

$T_1 = 280\text{K}$ and $T_2 = 370\text{K}$. Results show that the SSCC plate has the lowest but the SSFF plate has the highest deflection. The normal stress is the smallest in the SSSS plate while the biggest in the SSFF plate, among all boundary conditions considered. The maximum normal stresses occur near the points at a distance $h/2-s$ from the top and bottom surfaces of the plate. The normal stress distribution is seen to be un-symmetric despite the symmetric material compositional profile in the plate. This is because the temperature field induced by the heat conduction is un-symmetric along the plate thickness.

Figures 9 and 10 depict, respectively, the effects of thickness-to-length ratio h/a and aspect ratio a/b on the dimensionless deflection $w/h \times 10$ and normal stress $\sigma_x/E_m \times 10$ along the thickness direction of the plate with GPLs nonlinearly embedded in polymer matrix.

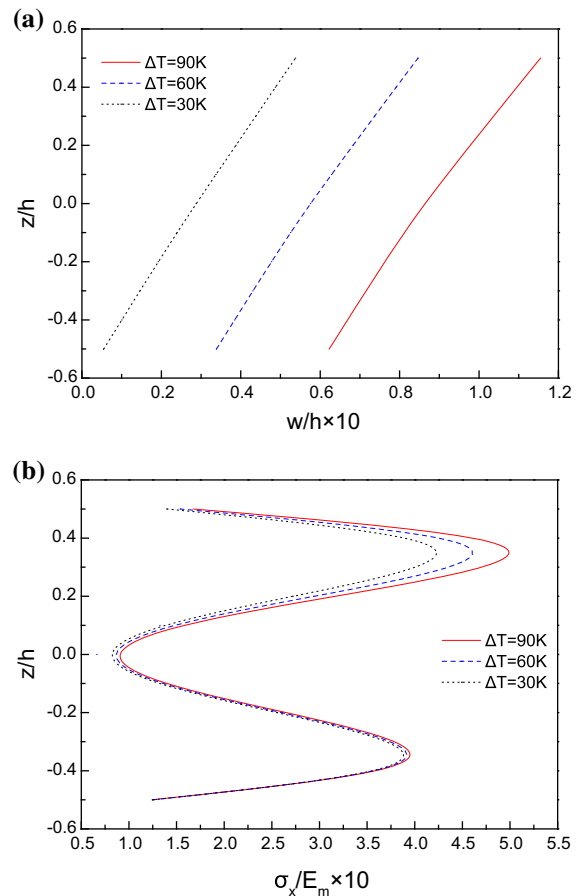


Fig. 7 Effect of temperature difference on the bending behavior of nanocomposite plate under heat conduction **a** dimensionless deflection; and **b** dimensionless normal stress

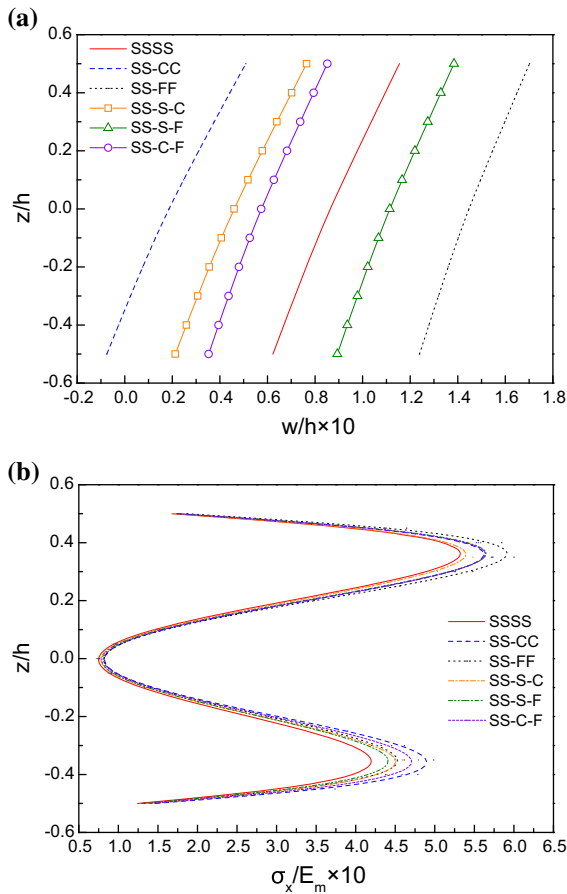


Fig. 8 Effect of boundary condition on the bending behavior of nanocomposite plate under heat conduction: through-thickness distribution of **a** dimensionless deflection; and **b** dimensionless normal stress

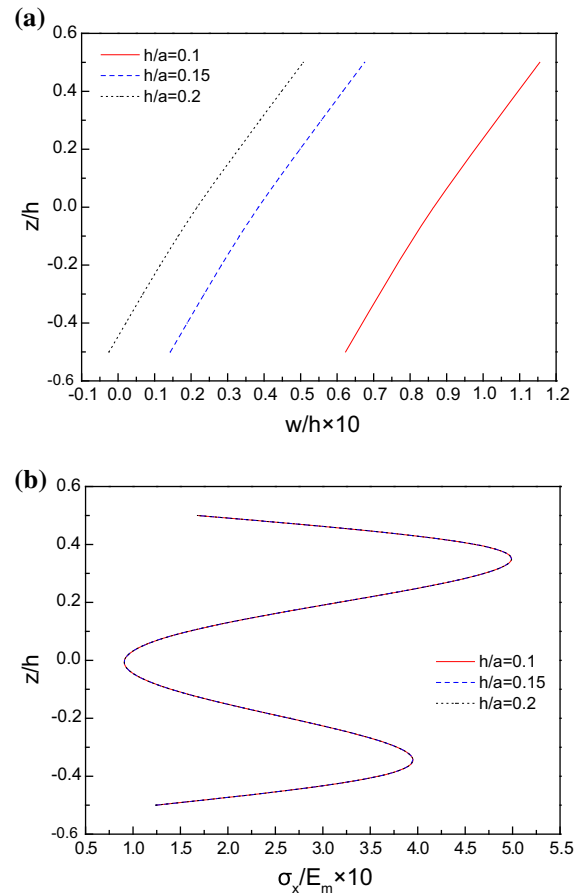


Fig. 9 Effect of h/a on the bending behavior of thermo-mechanically loaded nanocomposite plates: through-thickness distribution of **a** dimensionless deflection; and **b** dimensionless normal stress

The plate is subjected to a combined action of a uniform transverse load and heat conduction. As can be seen, a thinner plate with a bigger aspect ratio undergoes larger bending deformation induced by the same thermo-mechanical loading. The normal stress is not affected by the thickness-to-length ratio h/a . The aspect ratio, however, does influence both the deflection and normal stress considerably, as evidenced by the square plate having much less deflection and much higher normal stress than rectangular plates ($a/b = 2, 3$).

We now investigate the effects of GPL distribution pattern and weight fraction on the dimensionless shear stress of thermo-mechanically loaded plate. For this

purpose, the dimensionless shear stress results $\sigma_{xz}/E_m \times 10^6$ at $(x = a/4, y = b/4)$ along the plate thickness are compared in Fig. 11. The shear stress distribution is symmetric with its maximum value located at the mid-plane when GPLs are symmetrically dispersed but when GPLs are linearly distributed which is un-symmetric about the mid-plane, the peak value of the shear stress, which is bigger than those in other two GPL distributions, is found to be in the top side of the plate with the highest GPL concentration. With GPL weight fraction increasing from 0.33% to 1.0%, the maximum shear stress just slightly drops for nonlinear GPL distribution but increases much more significantly for linear GPL distribution. In the case of

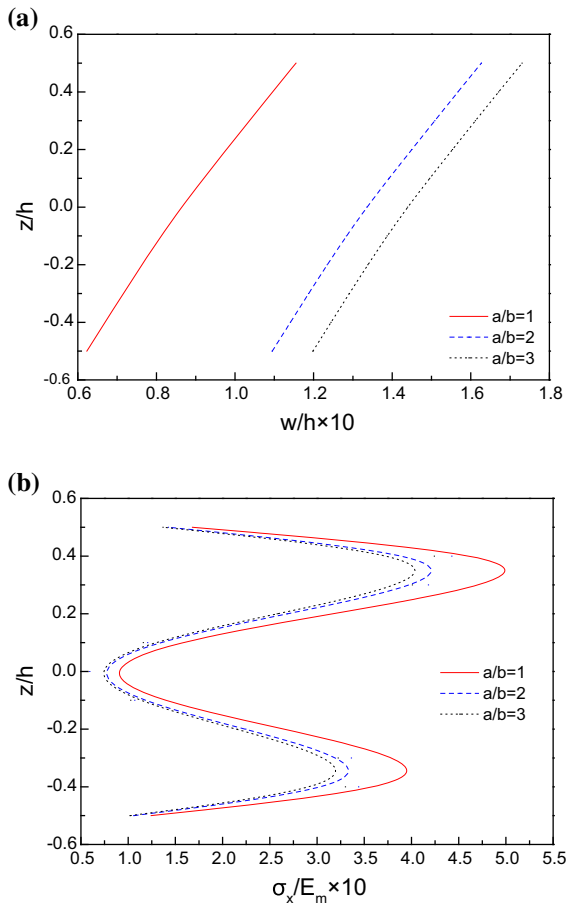


Fig. 10 Effect of a/b on the bending behavior of thermo-mechanically loaded nanocomposite plates: through-thickness distribution of **a** dimensionless deflection; and **b** dimensionless normal stress

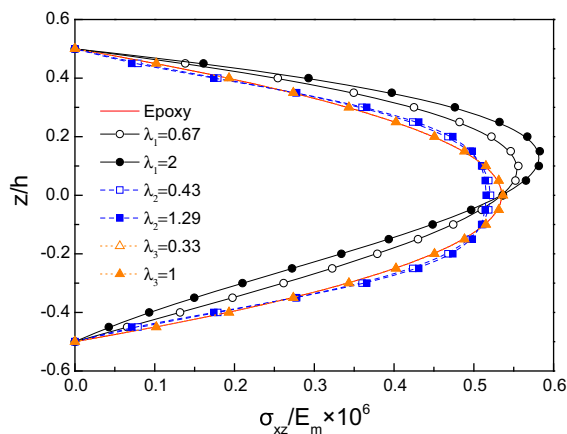


Fig. 11 Through-thickness distribution of dimensionless shear stress of the thermo-mechanically loaded nanocomposite plate

uniform GPL distribution, the through-thickness shear stress variation is almost identical to that of the pure epoxy plate. Consistent with the normal stress results in Fig. 5b, the nonlinear GPL distribution gives the lowest maximum shear stress among the three patterns considered.

5 Conclusions

Based on 3D elasticity theory and the generalized Mian and Spencer method, the thermo-elastic bending of functionally graded polymer nanocomposite rectangular plates reinforced with GPLs under the combined thermo-mechanical loads has been studied. Analytical solutions have been obtained for plates with two opposite edges simply supported. It is found that:

- Incorporating a small amount of GPLs into the matrix can lead to significantly reinforced plate stiffness.
- Compared to rectangular GPLs, squared GPLs with fewer single graphene layers offer better load transfer capability through bigger surface area which can most significantly strengthen the plate stiffness;
- Among the GPL distribution patterns considered, the nonlinear distribution pattern is preferred since it can yield the lowest bending deflection and the reduced maximum normal stress;
- A higher temperature difference ΔT results in greatly increased bending deflection and larger normal stress;
- With an increase in GPL weight fraction, the maximum shear stress slightly decreases for nonlinear GPL distribution pattern but significantly increases for linear GPL distribution pattern. The nonlinear GPL distribution also gives the lowest maximum shear stress among the three GPL patterns considered.

It should be pointed out that the derived analytical solutions exactly satisfy the equilibrium equations and the traction boundary conditions on both top and bottom surfaces of the plate. The boundary conditions on the cylindrical edge of the plate are approximately satisfied in the Saint-Venant sense. The present elasticity solutions can serve as a benchmark for simplified plate theories based analytical or numerical

solutions for the functionally graded graphene-reinforced composite plates.

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Appendix 1

$$\begin{aligned}
 g_i^n(z) &= \int_{-h/2}^z (z - \xi)^n \xi^i \frac{c_{13}(\xi)}{c_{33}(\xi)} d\xi, \quad (i, n = 0, 1), \\
 h_i^n(z) &= \int_{-h/2}^z (z - \xi)^n \xi^i \left[c_{11}(\xi) - \frac{c_{13}^2(\xi)}{c_{33}(\xi)} \right] d\xi, \quad (i, n = 0, 1), \\
 g_6(z) &= \int_{-h/2}^z c_{66}(\xi) d\xi, \quad f_3(z) = \int_{-h/2}^z c_{33}^{-1}(\xi) d\xi, \\
 f_4(z) &= \int_{-h/2}^z h_1^0(\xi) c_{44}^{-1}(\xi) d\xi, \quad f_5(z) = \int_{-h/2}^z h_0^0(\xi) c_{44}^{-1}(\xi) d\xi, \\
 f_6(z) &= \int_{-h/2}^z g_6(\xi) c_{44}^{-1}(\xi) d\xi, \\
 f_7(z) &= \int_{-h/2}^z c_{33}^{-1}(\xi) \{ h_0^1(\xi) + c_{13}(\xi) [f_5(\xi) \\
 &\quad - g_0^1(\xi) + (\xi + h/2)G_0 - F_0] \} d\xi, \\
 f_8(z) &= \int_{-h/2}^z c_{33}^{-1}(\xi) \{ h_1^1(\xi) + c_{13}(\xi) [f_4(\xi) \\
 &\quad - g_1^1(\xi) - (\xi + h/2)C_0 + B_0] \} d\xi, \\
 f_9(z) &= \int_{-h/2}^z [2c_{13}(\xi) c_{33}^{-1}(\xi) \alpha_1(\xi) + \alpha_3(\xi)] T(\xi) d\xi, \\
 \kappa_1 &= \frac{h_0^0(h/2)}{g_6(h/2)}, \quad \kappa_2 = -\frac{h_1^0(h/2)}{g_6(h/2)}, \quad \kappa_3 = -\kappa_2 \kappa_4 / \kappa_1, \\
 \kappa_4 &= \kappa_1 (q_2 - q_1) / [\kappa_2 h_0^1(h/2) + \kappa_1 h_1^1(h/2)], \\
 B_0 &= g_1^1(0) - h g_1^0(0) / 2 - f_4(0) - \kappa_2 f_6(0), \quad C_0 = -g_1^0(0), \\
 D_0 &= -\kappa_3 f_7(0) + \kappa_4 f_8(0) - f_9(0) + q_1 f_3(0), \\
 F_0 &= f_5(0) + h g_0^0(0) / 2 - g_0^1(0) - \kappa_1 f_6(0), \quad G_0 = g_0^0(0).
 \end{aligned}
 \tag{88}$$

Appendix 2

$$\begin{aligned}
 N_0 &= \int_{-h/2}^{h/2} \{ c_{13} D' - [(c_{11} + c_{12}) \alpha_1 + c_{13} \alpha_3] T \} dz, \\
 N_1 &= h_0^0(h/2), \\
 N_2 &= \int_{-h/2}^{h/2} (c_{12} - c_{13}^2 / c_{33}) dz, \quad N_3 = -h_1^0(h/2), \\
 N_4 &= - \int_{-h/2}^{h/2} (c_{12} - c_{13}^2 / c_{33}) z dz, \\
 N_5 &= \int_{-h/2}^{h/2} c_{11} F dz, \quad N_6 = \int_{-h/2}^{h/2} c_{12} F dz, \\
 N_7 &= \int_{-h/2}^{h/2} c_{11} B dz, \quad N_8 = \int_{-h/2}^{h/2} c_{12} B dz.
 \end{aligned}
 \tag{89}$$

$$\begin{aligned}
 M_0 &= \int_{-h/2}^{h/2} \{ c_{13} D' - [(c_{11} + c_{12}) \alpha_1 + c_{13} \alpha_3] T \} z dz, \\
 M_1 &= h_1^0(h/2) \\
 M_2 &= \int_{-h/2}^{h/2} (c_{12} - c_{13}^2 / c_{33}) z dz, \quad M_3 = -h_2^0(h/2), \\
 M_4 &= - \int_{-h/2}^{h/2} (c_{12} - c_{13}^2 / c_{33}) z^2 dz, \\
 M_5 &= \int_{-h/2}^{h/2} c_{11} z F dz, \quad M_6 = \int_{-h/2}^{h/2} c_{12} z F dz, \\
 M_7 &= \int_{-h/2}^{h/2} c_{11} z B dz, \quad M_8 = \int_{-h/2}^{h/2} c_{12} z B dz.
 \end{aligned}
 \tag{90}$$

$$\begin{aligned}
 T_1 &= (N_1 - N_2) / 2, \quad T_2 = (N_3 - N_4) / 2, \\
 T_3 &= (N_5 - N_6) / 2, \quad T_4 = (N_7 - N_8) / 2.
 \end{aligned}
 \tag{91}$$

$$\begin{aligned}
 S_1 &= (M_1 - M_2) / 2, \quad S_2 = (M_3 - M_4) / 2, \\
 S_3 &= (M_5 - M_6) / 2, \quad S_4 = (M_7 - M_8) / 2.
 \end{aligned}
 \tag{92}$$

$$\begin{aligned} Q_1 &= \int_{-h/2}^{h/2} c_{44}(F' + G)dz, \\ Q_2 &= \int_{-h/2}^{h/2} c_{44}(B' + C)dz. \end{aligned} \quad (93)$$

Appendix 3

$$\begin{aligned} \tilde{N}_{xm}(y) &= N_2 v'_m(y) - N_1 \alpha_m u_m(y) \\ &\quad - N_5 \alpha_m^2 [v'_m(y) - \alpha_m u_m(y)] \\ &\quad + N_6 [v'_m(y) - \alpha_m u_m(y)]'' + N_4 w''_m(y) \\ &\quad - N_3 \alpha_m^2 w_m(y) - N_7 \alpha_m^2 [w''_m(y) - \alpha_m^2 w_m(y)] \\ &\quad + N_8 [w''_m(y) - \alpha_m^2 w_m(y)]''. \end{aligned} \quad (94)$$

$$\begin{aligned} \tilde{N}_{ym}(y) &= N_1 v'_m(y) - N_2 \alpha_m u_m(y) \\ &\quad - N_6 \alpha_m^2 [v'_m(y) - \alpha_m u_m(y)] \\ &\quad + N_5 [v'_m(y) - \alpha_m u_m(y)]'' + N_3 w''_m(y) \\ &\quad - N_4 \alpha_m^2 w_m(y) - N_8 \alpha_m^2 [w''_m(y) - \alpha_m^2 w_m(y)] \\ &\quad + N_7 [w''_m(y) - \alpha_m^2 w_m(y)]''. \end{aligned} \quad (95)$$

$$\begin{aligned} \tilde{N}_{xym}(y) &= T_1 [u'_m(y) + \alpha_m v_m(y)] + 2T_2 \alpha_m w'_m(y) \\ &\quad + 2T_3 \alpha_m [v'_m(y) - \alpha_m u_m(y)]' \\ &\quad + 2T_4 \alpha_m [w''_m(y) - \alpha_m^2 w_m(y)]'. \end{aligned} \quad (96)$$

$$\begin{aligned} \tilde{M}_{xm}(y) &= M_2 v'_m(y) - M_1 \alpha_m u_m(y) - M_5 \alpha_m^2 [v'_m(y) \\ &\quad - \alpha_m u_m(y)] + M_6 [v'_m(y) - \alpha_m u_m(y)]'' \\ &\quad + M_4 w''_m(y) - M_3 \alpha_m^2 w_m(y) \\ &\quad - M_7 \alpha_m^2 [w''_m(y) - \alpha_m^2 w_m(y)] \\ &\quad + M_8 [w''_m(y) - \alpha_m^2 w_m(y)]''. \end{aligned} \quad (97)$$

$$\begin{aligned} \tilde{M}_{ym}(y) &= M_1 v'_m(y) - M_2 \alpha_m u_m(y) - M_6 \alpha_m^2 [v'_m(y) \\ &\quad - \alpha_m u_m(y)] + M_5 [v'_m(y) - \alpha_m u_m(y)]'' \\ &\quad + M_3 w''_m(y) - M_4 \alpha_m^2 w_m(y) \\ &\quad - M_8 \alpha_m^2 [w''_m(y) - \alpha_m^2 w_m(y)] \\ &\quad + M_7 [w''_m(y) - \alpha_m^2 w_m(y)]''. \end{aligned} \quad (98)$$

$$\begin{aligned} \tilde{M}_{xym}(y) &= S_1 [u'_m(y) + \alpha_m v_m(y)] + 2S_2 \alpha_m w'_m(y) \\ &\quad + 2S_3 \alpha_m [v'_m(y) - \alpha_m u_m(y)]' \\ &\quad + 2S_4 \alpha_m [w''_m(y) - \alpha_m^2 w_m(y)]'. \end{aligned} \quad (99)$$

$$\begin{aligned} Q_{xm}(y) &= Q_1 \alpha_m [v'_m(y) - \alpha_m u_m(y)] + Q_2 \alpha_m [w''_m(y) \\ &\quad - \alpha_m^2 w_m(y)]. \end{aligned} \quad (100)$$

$$\begin{aligned} Q_{ym}(y) &= Q_1 [v'_m(y) - \alpha_m u_m(y)]' + Q_2 [w''_m(y) \\ &\quad - \alpha_m^2 w_m(y)]'. \end{aligned} \quad (101)$$

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